ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** glow dimly	4.5 V is less than the rated 6 V.
2	**(b)** test with a tester; add salt to water to make it conduct	Dissolved salts supply ions.
3	**(c)** a conductor	Current passes through it.
4	**(c)** glow very bright and may fuse	Exceeds rating.
5	**(a)** tap water has dissolved salts	Dissolved ions conduct.
6	**(b)** an insulator	Current does not pass.
7	**(c)** series	Single path.
8	**(a)** copper conducts better	Brightness suggests current.
9	**(a)** both conduct, copper better	Brightness shows current.
10	**(a)** a single path for the current	The loop is broken.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Tungsten has a very high melting point (1); Copper would melt before glowing (1).	1+1
12	Metals are good conductors (1); They carry current to the appliance (1).	1+1
13	Prevent touching live parts (1); Avoid shocks (1).	1+1
14	Nichrome gets very hot when current passes (1); It does not melt or oxidise easily at high temperatures (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	6 x 1.5 = 9 V (1); Much higher than 3 V (1); Bulb may fuse (1).	1+1+1
16	They need a large current (1); Thin wiring may overheat (1); Risk of fire (1).	1+1+1
17	Switch off the supply (1); Dry hands (1); Use proper bulb (1).	1+1+1
	OR: Thin wire heats more (1); Melts first (1); Protects the circuit (1).	1+1+1
18	Iron would melt or burn (1); Tungsten has a high melting point (1); Bulb has inert gas (1).	1+1+1
	OR: One path (1); Same current (1); If one fails all stop (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	All bulbs go off (1); Only one path (1); Test each bulb with a tester (1); Replace the faulty one (1); Use a fresh bulb for check (1).	1+1+1+1+1
	OR: Marks show quality (1); Safer products (1); Fuse rating matches use (1); Do not use higher rating (1); Check before buying (1).	1+1+1+1+1
20	Switch off unused devices (1); Use LED bulbs (1); Use daylight (1); Do not overload (1); Service appliances (1).	1+1+1+1+1
	OR: Overloading (1); Short circuits (1); Damaged wires (1); Use ISI products (1); Use fuse or MCB (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** the nail is coated with paint or rust	Surface layer blocks contact.
2	**(d)** goes off	The loop is broken.
3	**(c)** its melting point is lower than that of the connecting wires	Low melting.
4	**(c)** touch it	Touching is unsafe.
5	**(b)** the reversed cell reduces the total voltage	Opposed cells cancel partly.
6	**(d)** the lemon juice conducts	Ions in the juice.
7	**(b)** to avoid contact and shock	Safety.
8	**(c)** a broken connecting wire	Open path.
9	**(d)** the bulb has inert gas or vacuum	No oxygen.
10	**(d)** exposed wire can shock	Insulation damage.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cells in series add their voltages (1); Higher voltage drives more current through the filament (1).	1+1
12	Plastic is an insulator (1); It protects the user from shocks (1).	1+1
13	15 A fuse will not protect thin wires (1); They may overheat (1).	1+1
14	It can cut off the current to the appliance (1); It controls the appliance (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	9 / 1.5 = 6 (1); Six cells in series (1); + to - connections (1).	1+1+1
16	Switch off appliances (1); Replace the fuse with the same rating (1); Avoid overloading (1).	1+1+1
17	Chemicals may leak (1); Damage to the toy (1); Cell is saved (1).	1+1+1
	OR: Opposing cells reduce voltage (1); Little or no current (1); Filament does not heat (1).	1+1+1
18	Use less energy (1); Less fuel burnt (1); Fewer emissions (1).	1+1+1
	OR: Saves electrical energy (1); Reduces bills (1); Saves resources (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Same cells, wires of different length (1); Fix a small piece of wax on each wire (1); Observe how soon the wax melts (1); Compare (1); Safe handling (1).	1+1+1+1+1
	OR: Heating plate: metal alloy (1); Body: plastic insulator (1); Reason: shock safety (1); Light weight (1); Heat resistant plastic (1).	1+1+1+1+1
20	Insulated wiring (1); Fuse and MCB (1); Switches in good condition (1); Dry hands (1); Switch off and call help (1).	1+1+1+1+1
	OR: Light at night (1); Machines run (1); Communication (1); Rules prevent accidents (1); Careful use (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** iron nail, rubber, aluminium foil	Conductor, insulator, conductor.
2	**(a)** heating effect	Wire melts by heat.
3	**(d)** a sealed metal sheath with insulating powder	Safe design.
4	**(c)** it can carry more current without overheating	Thick wires heat less.
5	**(c)** a long wire fits in a small space	Compact design.
6	**(a)** switching to LED bulbs	Efficient.
7	**(d)** water conducts and can give a shock	Safety.
8	**(a)** connecting many appliances to one socket	Large current.
9	**(c)** open	No complete path.
10	**(c)** replacing a blown fuse of the same rating after switching off the supply	Safe practice.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Current flows quickly through the circuit (1); All parts conduct (1).	1+1
12	Positive of one to negative of next (1); 3 x 1.5 V = 4.5 V (1).	1+1
13	Thin wires overheat (1); Safe current carrying (1).	1+1
14	Conduct and heat (1); Can be shaped as coils (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Large current (1); Wires heat (1); Insulation burns (1).	1+1+1
16	Distilled water has no dissolved salts (1); Tap water has dissolved salts (1); So it conducts (1).	1+1+1
17	Lighter (1); Cheaper (1); Good conductor (1).	1+1+1
	OR: Dip tester ends (1); Bulb glows for conducting liquids (1); Lemon and salt solution conduct, sugar does not (1).	1+1+1
18	More current more heat (1); Heat increases with current (1); Fuse chosen accordingly (1).	1+1+1
	OR: Overheating (1); Damage to holder and wiring (1); Fire (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	3 cells in series (1); Total 4.5 V (1); Bulb glows normally (1); 2 cells give 3 V (1); Bulb dim (1).	1+1+1+1+1
	OR: Used in iron, heater (1); Easy to control (1); Energy wasted in devices (1); Wire heating hazard (1); Use fuse (1).	1+1+1+1+1
20	Tester shows conduction (1); Metals light the bulb (1); Poor conductors give a weak current (1); Needs good contacts (1); Cannot measure resistance (1).	1+1+1+1+1
	OR: In series with the appliance (1); Not across the appliance (1); Current passes through it (1); Melts on overload (1); Appliance isolated (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.